

HSBI — Bielefeld University of Applied Sciences and Arts

Quantitative Research Methods

Summer Semester 2026 · Examiner: Prof. Dr. Christoph Weisser

Final Mock Examination (Lectures 1–12)

Comprehensive coverage of ISLP Chapters 2–8, 10 and 13,
with emphasis on Chapters 7 (splines/GAMs), 8 (trees), 10 (deep learning) and 13 (multiple testing)

Duration: 120 minutes

Total credit: 120 points

Allowed aids: non-programmable calculator; one handwritten A4 sheet of notes (both sides)

Name: _____

Matriculation no.: _____

Problem	P1	P2	P3	P4	P5	P6	P7	Total
Maximum points	16	8	20	16	20	20	20	120
Achieved points								

Instructions

- Justify all answers. Numerical results without a derivation or a brief reasoning receive no credit.
- Round final numerical results to 3 decimal places unless stated otherwise. Small deviations caused by carrying rounded intermediate results are accepted.
- All numerical constants you need (powers of e , logarithms, $(0.95)^{20}$, t -quantiles) are provided in the problems.
- The number of points per problem roughly equals the recommended working time in minutes.

Problem 1: Moving beyond linearity — splines

(16 points)

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

- (a) [4 P] A *cubic spline* is a piecewise cubic polynomial that is continuous and has continuous first and second derivatives at each knot. How many degrees of freedom (free parameters, counting the intercept) does a cubic spline with K interior knots use? Justify your answer by counting the parameters of the polynomial pieces and the constraints at the knots, and evaluate the result for $K = 4$.
- (b) [4 P] How many degrees of freedom does a *natural* cubic spline with the same K knots use (again evaluate for $K = 4$)? Which additional constraint does a natural spline impose, in which region of the predictor space does it act, and why is this constraint useful in practice?
- (c) [4 P] A *smoothing spline* is the function g that minimises

$$\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt.$$

Explain the role of the two terms, and describe the behaviour of the solution $\hat{g}$ in the two limiting cases $\lambda \rightarrow 0$ and $\lambda \rightarrow \infty$.

- (d) **[4 P]** Write down the *truncated-power basis* representation of a cubic spline with a single knot at ξ , define the truncated-power basis function explicitly, and explain in one sentence why adding this basis function does not destroy the smoothness of the fit at ξ .

Problem 2: Generalised additive models

(8 points)

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

- (a) **[3 P]** Write down the general form of a *generalised additive model* (GAM) for a quantitative response Y and predictors $X_1, \dots, X_p$, and explain in one sentence how it extends multiple linear regression.
- (b) **[3 P]** State one important *advantage* of GAMs that concerns the interpretation of the fitted functions f_j , and the main structural *limitation* of GAMs — including how this limitation can be repaired.
- (c) **[2 P]** Describe in one or two sentences how the *backfitting* algorithm fits a GAM.

Problem 3: Regression and classification trees by hand

(20 points)

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

An online platform records for $n = 6$ products the advertising budget x (in 1,000 EUR) and the weekly sales y (in 1,000 units):

Product i	1	2	3	4	5	6
Budget x_i	1	2	3	4	5	6
Sales y_i	2	4	3	3	8	10

A regression tree with a *single* split at cutpoint s partitions the data into $R_1 = \{i : x_i < s\}$ and $R_2 = \{i : x_i \geq s\}$ and predicts the region mean $\hat{y}_{R_m}$ in each region.

- (a) **[8 P]** For each of the two candidate cutpoints $s = 2.5$ and $s = 4.5$, determine the two regions, the region means and the resulting total residual sum of squares $\text{RSS}(s) = \sum_{m=1}^2 \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2$. Which cutpoint does recursive binary splitting choose, and why?
- (b) **[2 P]** Using the split chosen in part (a), namely $s = 4.5$ (so $R_1 = \{i : x_i < 4.5\}$ with mean $\hat{y}_{R_1} = 3$ and $R_2 = \{i : x_i \geq 4.5\}$ with mean $\hat{y}_{R_2} = 9$), predict the weekly sales for a new product with advertising budget $x = 5$.
- (c) **[6 P]** A *classification* tree for a different task has a terminal node containing 8 training observations: 6 of class “Yes” and 2 of class “No”. Compute the class proportions, the Gini index $G = \sum_k \hat{p}_k(1 - \hat{p}_k)$ and the entropy $D = -\sum_k \hat{p}_k \ln \hat{p}_k$ of this node (use $\ln 0.75 = -0.2877$ and $\ln 0.25 = -1.3863$). Which class does the node predict?
- (d) **[4 P]** In *cost-complexity pruning* one minimises, for a subtree $T \subseteq T_0$ of the large tree T_0 ,

$$\sum_{m=1}^{|T|} \sum_{i: x_i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|.$$

Explain the role of the tuning parameter α : what happens for $\alpha = 0$, what happens as α increases, and how is α chosen in practice?

Problem 4: Ensembles — bagging, random forests, boosting**(16 points)**

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

- (a) **[6 P]** Let $\hat{f}_1(x), \dots, \hat{f}_B(x)$ be B identically distributed (but not independent) tree predictions at a fixed point x , each with variance σ^2 and with pairwise correlation ρ . The variance of the bagged average is

$$\text{Var}\left(\frac{1}{B} \sum_{b=1}^B \hat{f}_b(x)\right) = \rho \sigma^2 + \frac{1-\rho}{B} \sigma^2.$$

- (i) Explain briefly why this formula shows that bagging reduces variance, and which term limits the reduction. (ii) Evaluate the variance for $B = 100$, $\rho = 0.6$ and $\sigma^2 = 4$, and compare it with the variance of a single tree. (iii) What value does the variance approach as $B \rightarrow \infty$?
- (b) **[4 P]** How do *random forests* modify bagging at each split, and what is the recommended subset size m for classification as a function of p (evaluate for $p = 25$)? Explain — with reference to the ensemble-variance formula $\text{Var} = \rho \sigma^2 + \frac{1-\rho}{B} \sigma^2$ from part (a) — why this modification improves on bagging.
- (c) **[4 P]** A data scientist tunes a *boosting* model for regression trees with the following `scikit-learn` call:

```
GradientBoostingRegressor(n_estimators=5000,  
                           learning_rate=0.01, max_depth=1)
```

These three constructor arguments are exactly the three tuning parameters of boosting for regression trees. State in one sentence each what `n_estimators`, `learning_rate` and `max_depth` control. What is the “slow learning” idea behind boosting, and how is it reflected in the values chosen here?

- (d) **[2 P]** What does the *out-of-bag* (OOB) error of a bagged model estimate, and why is it available essentially for free?

Problem 5: Neural networks by hand**(20 points)**

Lecture slides: Chapter 10 — Deep Learning (Lecture 11).

Consider a tiny feed-forward network for a quantitative response with input $x = (x_1, x_2) = (1, 2)$, one hidden layer with two ReLU units ($g(z) = \max\{0, z\}$) and a linear output unit:

$$A_1 = g(-1 + 1x_1 + 2x_2), \quad A_2 = g(-3 - 2x_1 + 1x_2), \quad f(x) = 1 + 2A_1 + 4A_2.$$

- (a) **[8 P]** Perform the forward pass step by step: compute both pre-activations, both activations A_1, A_2 and the network output $f(x)$. Comment briefly on what the ReLU did to the second unit and why such non-linear activations are essential.
- (b) **[6 P]** A classification network with $K = 3$ classes produces the logits (output-layer pre-activations) $z = (z_1, z_2, z_3) = (1.0, 0.0, -1.0)$ for one observation whose true class is class 1. Compute all three softmax probabilities and the cross-entropy loss $-\ln \hat{p}_{\text{true}}$. (Use $e^1 = 2.718$, $e^0 = 1.000$, $e^{-1} = 0.368$ and $\ln 4.086 = 1.408$.)
- (c) **[4 P]** Count the parameters of a fully connected network with $p = 4$ inputs, one hidden layer with $k = 3$ units and an output layer with $K = 2$ units, where every hidden and output unit has a bias. Show the count separately for the two weight layers.

- (d) **[2 P]** A convolutional layer receives a 32×32 image and uses filters of size $F = 5$ with padding $P = 2$ and stride $S = 1$. Compute the spatial size of the output feature map from $(W - F + 2P)/S + 1$.

Problem 6: Multiple testing

(20 points)

Lecture slides: Chapter 13 — Multiple Testing (Lecture 12).

- (a) **[4 P]** A lab performs $m = 20$ independent hypothesis tests, each at level $\alpha = 0.05$, and suppose all 20 null hypotheses are true. Define the family-wise error rate (FWER) and compute it for this situation (use $(0.95)^{20} = 0.3585$). Interpret the result in one sentence.
- (b) **[7 P]** A research group screens $m = 6$ candidate biomarkers; the ordered p -values are

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.004	0.011	0.030	0.045	0.230

Control the FWER at $\alpha = 0.05$ (i) with *Bonferroni* and (ii) with the *Holm step-down* procedure. Give the relevant thresholds, state for each method exactly which hypotheses are rejected, and explain why Holm can never reject fewer hypotheses than Bonferroni.

- (c) **[6 P]** Apply the *Benjamini-Hochberg* (BH) procedure with FDR level $q = 0.05$ to the same six ordered p -values from part (b), namely $p_{(1)}, \dots, p_{(6)} = 0.001, 0.004, 0.011, 0.030, 0.045, 0.230$: give the thresholds jq/m , determine which hypotheses are rejected, and state precisely what “FDR control at $q = 0.05$ ” guarantees for the resulting set of rejections.
- (d) **[3 P]** A genomics group screens 20,000 genes in an *exploratory* study; the most promising hits will be re-tested in an independent follow-up experiment. Should they control the FWER or the FDR? Justify your recommendation.

Problem 7: Cumulative essentials (Chapters 2–6)

(20 points)

Lecture slides: Chapters 2–6 (Lectures 1–8).

- (a) **[4 P]** True or false — justify briefly: “A more flexible statistical learning method always yields a lower *test* error than a less flexible one.”
- (b) **[4 P]** A bank fits a logistic regression for loan default in `statsmodels`, $\log\left(\frac{p(x)}{1-p(x)}\right) = \beta_0 + \beta_1 \text{missed_payments}$, where `missed_payments` is the number of missed payments x . The call `sm.Logit(default, X).fit()` produces:

	coef	std err	z	$P > z $
<code>const</code>	−3.5000	0.420	−8.333	0.000
<code>missed_payments</code>	0.7000	0.150	4.667	0.000

Using the estimated coefficient of `missed_payments`, interpret the effect of *one additional missed payment* on the *odds* of default (use $e^{0.7} = 2.014$). Why can the effect on the *probability* of default not be stated as a single number?

- (c) **[4 P]** Give one reason to prefer k -fold cross-validation with $k = 5$ or 10 over $LOOCV$, and one reason to prefer it over a *single validation split*.
- (d) **[2 P]** You have $p = 2,000$ predictors but expect only a handful to be truly related to the response. Ridge or lasso — which penalty is more appropriate, and why?
- (e) **[6 P]** Match each scenario to exactly one method from $\{\textit{linear regression}, \textit{lasso}, \textit{logistic regression}, \textit{random forest}\}$ (each method used exactly once) and justify each choice in one sentence:
- (i) Estimate how much an additional square metre adds to the price of an apartment, with a confidence interval, from $n = 2,000$ sales.
 - (ii) Predict a continuous drug response from $p = 5,000$ gene-expression measurements for $n = 100$ patients; clinicians want a short list of relevant genes.
 - (iii) Predict whether a loan applicant will default (yes/no) and report to the regulator how each applicant characteristic changes the odds of default.
 - (iv) Predict machine failures from hundreds of sensor variables with strong non-linear effects and interactions; only predictive accuracy matters.

End of the examination — good luck!